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Sub-cohort dynamics of 0-group plaice, *Pleuronectes platessa* L., in the Northern Irish Sea: Settlement, growth and mortality

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ABSTRACT

The settlement, growth and mortality of plaice (*Pleuronectes platessa* L.) were examined on a small nursery ground (Port Erin Bay, Isle of Man) in the Irish Sea in 1993 and between 1996 and 2000. The timing, duration and strength of juvenile settlement varied between years and were positively correlated with the duration and quantity of egg production. Otolith increment counts were used to determine the age and metamorphosis and/or settlement dates of fish and to compare settlement patterns inferred from catch data with those inferred from age data. The catch data suggested two 'pulses' of settlement in Port Erin Bay whereas the otolith age data indicated three main settlement events. Over the years the first sub-cohort was generally the largest and overall this sub-cohort suffered the highest mortality. This first sub-cohort may have a "high risk" strategy and swamp potential predators early in the settlement period, with the result that the second sub-cohort generally has faster growth rate and lower mortality. A release of predatory pressure on the second sub-cohort could be a cause of large year classes in plaice populations. Growth rates were lowest for fish in the first sub-cohort, likely reflecting density-dependent effects and less optimal environmental conditions early in the year. The variations in instantaneous mortality rate between sub-cohorts, as well as inter-annually within sub-cohorts, illustrate the complex dynamics in the structure of these juvenile plaice populations on the nursery grounds. Global environment change effects are visible in the wider Irish Sea plaice population dynamics, with temperature dependent nursery ground processes as one of the contributing mechanisms.

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1. Introduction

The dynamics of fish nursery grounds are an integral part of the recruitment process in fish populations. Nursery grounds can be defined generally as habitats that enhance the growth and survival of juveniles (Gibson, 1994), or more specifically as habitats that make a greater than average contribution to the recruitment of the adult population compared to other juvenile habitats (Beck et al., 2001). In species with broadcast spawning the nursery areas are annually populated by new individuals, often involving a pelagic drift phase and a settlement period onto the nursery grounds (Able et al., 2005). There are annual variations in both the overall numbers of individuals arriving on a nursery ground and the settlement patterns or timing (Hyder and Nash, 1998). These aspects of settlement both affect the structure of the juvenile population and influence the nursery ground dynamics. Flatfish in general and plaice (*Pleuronectes platessa*) in particular have been well studied, moving from a pelagic larval phase to a demersal or benthic existence at settlement (Beverton, 1995; Geffen et al., 2007), and the nurseries are in relatively shallow water (Wimpenny, 1953), thus making the logistics of undertaking research much easier.

The plaice is a commercially exploited flatfish species common in northern European waters. Irish Sea plaice spawn between February and April (Simpson, 1959) and the pelagic stage lasts 40–60 days (Russell, 1976; Armstrong et al., 2001; Fox et al., 2007). Newly metamorphosed plaice are caught in shallow water sandy bays or on tidal flats beginning in late spring (van der Veer, 1986). Plaice then stay on the shallow nursery grounds until the late autumn to early winter (Colman, 1966; Riley and Corlett, 1966; Lockwood, 1974; Nash et al., 1994a,b), generally moving in to deeper water as they get larger (Gibson et al., 2002) although some individuals may overwinter in shallow water (Nash et al., 1992, 1994a). The pattern of settlement is not continuous; there are often pulses where a large number of individuals arrive in a short space of time. These peaks in settlement, or sub-cohorts (Al-Hossaini et al., 1989), are features of many flatfish populations (Rauck, 1974; Hovenkamp, 1991; Allen et al., 2008) but have not been detected in all populations (e.g. Karakiri et al., 1991; Berghahn et al., 1995; Amara and Paul, 2003). These waves of newly-settling juveniles may be caused by periodicity in spawning or hatching pulses (Allen et al., 2008), or may be the result of differential transport from different spawning locations (Al-Hossaini et al., 1989) or selective tidal transport (Creutzberg et al., 1978; Rijnsdorp et al., 1985). Previous work in the Irish Sea (Red Wharf Bay, North Wales, Irish Sea) identified three groups of settlers or sub-cohorts which arrive in Irish Sea nursery grounds (Al-Hossaini et al., 1989). The dynamics of each of these sub-cohorts can be very different. Al-Hossaini et al. (1989) and Modin and Pihl (1994) (working on a Swedish nursery ground) suggested that the second sub-cohorts have a higher growth rate than the first and third sub-cohorts, while later sub-cohorts have extremely low growth rates. This has been associated with increased temperatures as the summer progresses. There may also be differences in the mortality rates experienced by the different sub-cohorts (Al-Hossaini et al., 1989).

Information from otolith analyses has made a significant contribution to our understanding of nursery ground processes. The otolith increments formed during the larval and early juvenile stages are particularly amenable to analysis, providing information about hatching and settlement dates, relative growth rates, and sometimes, duration of different life history stages. There are structural landmarks in the otolith that can be related to developmental events such as hatching or metamorphosis, or to behavioural changes such as settling (Geffen et al., 2003). Increment counts between hatching and metamorphosis or settlement provide data on the duration of the larval period (van der Veer et al., 2000). Counts of increments and increment widths can be used to determine settlement dates and subsequent growth on different nursery grounds (Sogard and Able, 1992; Berghahn et al., 1995), allowing an evaluation of nursery ground quality in some instances.

The formation, growth, and timing of structural landmarks in juvenile plaice otoliths are well documented. Growth increments are formed daily beginning at hatching during the larval stage (Karakiri and von Westernhagen, 1989) and also after metamorphosis (Alhossaini and Pitcher, 1988). The first increment at hatching is distinct (Karakiri and von Westernhagen, 1989), and metamorphosis is also marked by wide and distinct increments and the initiation of accessory growth centres (Modin et al., 1996; Fox et al., 2007). Tidal cycles induce a regular pattern of "checks" to the increments formed after settlement (Geffen and Nash, 1995) similar to patterns observed in other juvenile flatfish (Sogard, 1991). The types of data most commonly extracted from juvenile plaice otoliths are the hatching date and larval stage duration (Hovenkamp, 1991; van der Veer and Witte, 1999; van der Veer et al., 2000; Fox et al., 2007), and the settlement dates and subsequent growth (Al-Hossaini et al., 1989; Modin and Pihl, 1994).

We examined the dynamics of juvenile plaice sub-cohorts arriving on a small Irish Sea nursery ground (Port Erin Bay, Isle of Man), using otolith increment counts to determine the age and settlement dates of the fish and to assign them to sub-cohorts. The effect of tidal cycle and lunar cycle on the settlement pattern was investigated. Growth and mortality rates of the sub-cohorts were also calculated and compared using an intensive sampling regime during a single year (1993). From the long-term nursery ground studies of Port Erin Bay (1989–2003), a sub-set of the data (1996–2000) was then analysed to reveal larger trends in settlement and mortality patterns on this typical flatfish nursery ground. These results are put in to the context of shifts in environmental regimes and global climate by identifying trends in Irish Sea wide recruitment dynamics, based on a 40 year time series of plaice spawning population size and the number of one year old fish (ICES, 2008).

2. Methods

2.1. Study site

Port Erin Bay (54° 20′ N, 4° 10′ W) is a small sandy bay (393,000 m^2) (Fig. 1) with a tidal excursion on spring tides of approximately 6.7 m,

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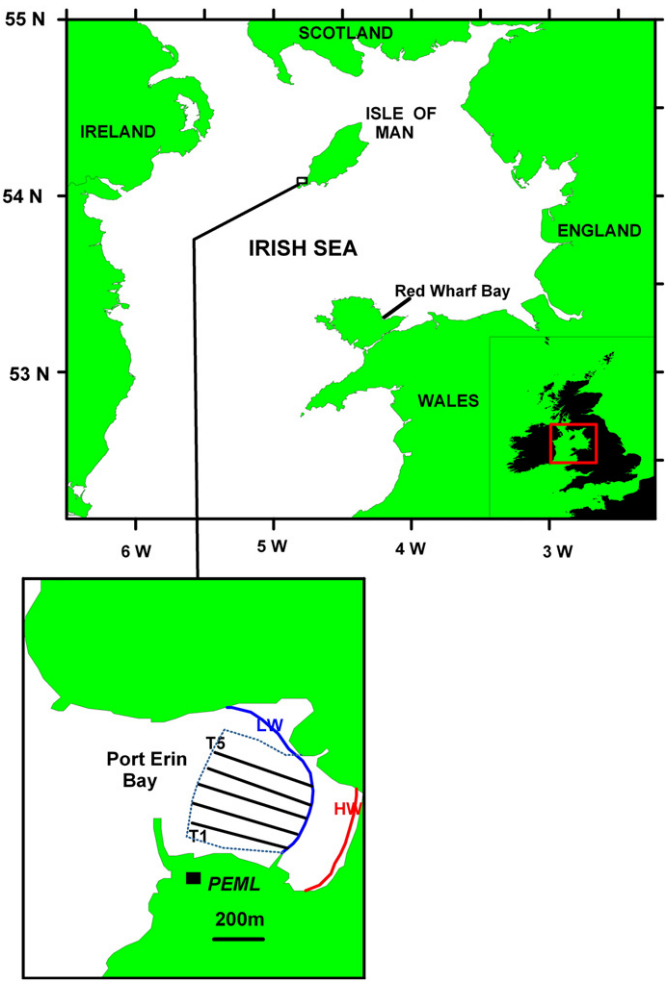

Fig. 1. Map of Port Erin Bay (54° 20′ N, 4° 10′ W), Isle of Man, Irish Sea with positions of trawl transects (T1–5). Push net sampling was at the low water mark (LW). The intertidal region is between LW and HW. PEML = Port Erin Marine Laboratory.

and is situated on the southwest side of the Isle of Man in the Irish Sea. The bay is a well-known nursery ground for juvenile plaice, and its populations there have been studied since the 1950s (Bowers, 1963; Jones and Kain, 1964; Riley and Holford, 1965; Riley and Corlett, 1966; Nash et al., 1992, 1994a,b; Hyder and Nash, 1998).

2.2. Sampling

Regular sampling of juvenile plaice in Port Erin Bay was conducted during the nursery season from 1989 to 2003. In 1993 an intensive sampling programme, including otolith analysis, was undertaken to identify pulses of settlement and follow the fate of individuals from different sub-cohorts. From 1996 to 2000, otolith analysis was included in the protocol, which allowed us to identify sub-cohorts and study differences in their settlement, growth, and mortality patterns. These two sampling programmes provided in-depth coverage of sub-cohort dynamics at a fine temporal scale and data on long-term patterns of settlement.

During 1993, the timing of juvenile plaice settlement in Port Erin Bay was established by sampling from early April until mid-July in shallow water (0.3–1 m) with a 1.5 m beam-trawl (modified Riley push net, Riley, 1973, 3 mm mesh throughout) three times weekly at low water, the period when the fish were most concentrated in the shallow water (Riley and Holford, 1965; Gibson et al., 1998, 2002). In addition, the whole of Port Erin Bay was surveyed by beam trawl between April and November. Juvenile flatfish were sampled along five parallel transects, 171–207 m long, running at right angles to the beach from 0.5 m to 9 m water depth (Fig. 1). Trawling speed was approximately 35 m min^{-1} to maximise the catch rate (Riley and Corlett, 1966). From April until July a 1.5 m beam-trawl with 5 mm stretched mesh throughout was used, and from July to November a 2 m beam-trawl with stretched mesh of 9 mm in the body and 5 mm in the codend was used. Efficiencies of the nets were considered to be 20% for the modified push net and 30% for the 2 m beam trawl (Nash and Geffen, 2000). Beam trawl sampling intensity was highest between April and the end of June (2 times per week). From June to the end of September sampling decreased to once per week, and biweekly samples were collected from October onwards. The exact sampling schedule varied with weather and sea conditions, but was always carried out within two hours of high water.

From April to July in 1996–2003, the newly settled and young fish in shallow water were sampled with a 0.33 m wide D-net with 1 mm mesh (see Nash and Geffen, 1999) at approximately weekly intervals at low tide. In general five trawls were taken parallel to the beach in shallow water over a distance of 30 m on each sample date.

All fish were returned to the laboratory, counted, measured to the nearest 1 mm, and then preserved in 70% ethanol for later extraction of otoliths.

2.3. Additional data sources

Sea-surface temperature data for Port Erin Bay were collected daily and supplied by T. Shammon (Port Erin Marine Laboratory, Isle of Man). The time and height of the tidal cycle were extracted from published tide tables (Laver's Liverpool and Irish Sea Tide Tables for 1993 and 1996 to 2000, www.laverpublishing.com).

The annual stage 1 plaice egg production for the west side of the Isle of Man was obtained from Nash and Geffen (1999), adding information from 1999 and 2000 to the previously published data. Data on the abundance of shrimp (*Crangon crangon*) and crabs (*Carcinus maenas*, *Liocarcinus puber*, *L. depurator*, and *L. holsatus*) were obtained from the regular beam trawl sampling in Port Erin Bay as reported in Nash and Geffen (2000), with the addition of data for 1999 and 2000. Data on the abundance of predatory fish in Port Erin Bay were obtained from beach seine sampling in 1996 and 1997 (de la Rosa, 1999). The protocols for the beach seine sampling and the description of the net are reported in Nash et al. (1994b).

2.4. Otolith microstructure analysis

The sagittal otoliths were removed from juvenile plaice and rinsed in hypochlorite (bleach) to dissolve any adhering tissue. The right otoliths were mounted on microscope slides using thermoplastic resin (Crystal Bond, melting at 40–60 °C) and ground to the core from both sides with carborundum paper (30, 9, and 1 μm). Primary increments were counted from the position of the first accessory (secondary) growth centre to the otolith edge at either 100× or 200× magnification (Fox et al., 2007).

Secondary growth centres on the otoliths mark the end of metamorphosis in juvenile plaice (see Alhossaini and Pitcher, 1988; Karakiri and von Westernhagen, 1989; Modin et al., 1996; Geffen et al., 2003), specifically at the completion of stage 5 (Modin et al., 1996). Although newly metamorphosed plaice do spend some time in the water column, we assumed that the presence of secondary growth centres also coincided with transition to the bottom. Plaice may settle in both deep and shallow water, and we assumed that the post-metamorphosis increment counts represented the number of days of post-settlement. Thus, fish with otoliths presenting zero increments outside the position of the first accessory growth centre were assumed to have settled in shallow water in the bay on the date of capture.

Data for analysis of detailed sub-cohort dynamics was obtained by examining the otoliths from a length-stratified subsample of the 1993

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beam trawl samples (n = 319). Each otolith was counted at least twice, on separate days. Reader error (Beamish and Fournier, 1981) was 1.5%.

To analyse the long-term patterns of settlement and sub-cohort dynamics, otoliths were dissected from randomly selected sub-samples of juvenile fish (sampled 1996–2000). Between 49 and 129 fish were sub-sampled for each year. Reader error was estimated from 57 otoliths read 5 times each, on separate dates, and the standard deviation of the results was averaged across the whole sample.

2.5. Data analysis

2.5.1. Settlement

The estimation of densities, abundances and numbers of settling fish was estimated following Nash and Geffen (1999). The dates of settlement were calculated for each individual based on the age post-metamorphosis. Individuals were grouped into three-day age classes, and the sub-cohorts of settling plaice were identified from the post-metamorphosis age frequency distribution at each sample date. The densities settling on each date were accumulated across the range of sample dates for annual totals. Older fish were included so as to give an overview of the settlement pattern, although the part of the population they represented would have sustained some mortality.

The catch data from the beam trawls and push net were analysed separately to compare the settlement patterns revealed by the different sampling regimes. The beam trawl sampled a large area at high tide, whereas the push net concentrated on the shallowest water at low tide. The size of the settling 0-group population (defined as fish in the size range 12–17 mm in total length) was estimated from beam trawl samples, taking the mean population estimate from the five transects. The standard deviation of the population estimate was the standard deviation of the averaged population size.

The settlement patterns revealed by the catch data and the otolith increment data were compared and tested against the lunar cycle (tidal heights), in order to assess the impact of the tidal cycle on the formation of the sub-cohorts.

2.5.2. Growth

Growth rates were calculated over the cohort settlement period and also for individual sub-cohorts in each year, by linear regression of length on age. The growth rates were compared between sub-cohorts and between years with respect to temperature and juvenile densities. Differences in growth rates were tested with analysis of covariance (Systat Inc.) followed by the Tukey test for pair-wise comparisons.

2.5.3. Mortality

Daily instantaneous mortality rates (including early mortality rate, EMR, Nash and Geffen, 2000) were calculated from the linear regression of $\mathrm{Log_e}$ density on age, for each year (van der Veer and Bergman, 1987). Individual sub-cohort daily instantaneous mortality rates were calculated in the same way but by isolating the fish belonging to each sub-cohort and taking them as a proportion of the total numbers of settlers in the respective sub cohort. The date on which the latest settling fish, according to this survey, reached 30 mm was calculated from the daily growth rate of the third sub-cohort.

2.5.4. Time series of the Irish Sea plaice population

The Irish Sea plaice population data from ICES (2008) were used to test the relationships between survival and thermal regimes during the first year of life. Spawning Stock Biomass (SSB) was taken directly from the assessment reports and egg production was calculated from the mean weights-at-age and population abundances as detailed by Nash (1998). The reports give the recruitment as numbers of individuals at 2 years old, so the numbers of one year olds were back-calculated assuming a natural mortality of 0.2 year^{-1}. The thermal regime was estimated as a three year running average of the daily Port Erin Bay sea-surface water temperature series (1963 to 2006) (data provided by T. Shammon, Port Erin Marine Laboratory, Port Erin, Isle of Man).

3. Results

3.1. Settlement

3.1.1. Comparisons between settlement and sub-cohort patterns

Two settlement peaks were identified directly from the beam trawl catch data (Fig. 2a). Settlement started at the end of April and increased rapidly to a peak at the beginning of May. A second, slightly smaller peak was seen at the end of May and settlement decreased during June and ceased completely by the beginning of July. Both peaks in settlement appeared to be associated with spring tides i.e. at maximum tidal amplitude.

The push net samples also showed two settlement peaks based on the abundances of newly-settled fish in < 1 m water depth (Fig. 2b). The peaks in the arrival of newly settled fish were more clearly separated in the push net samples than in the beam trawl transect

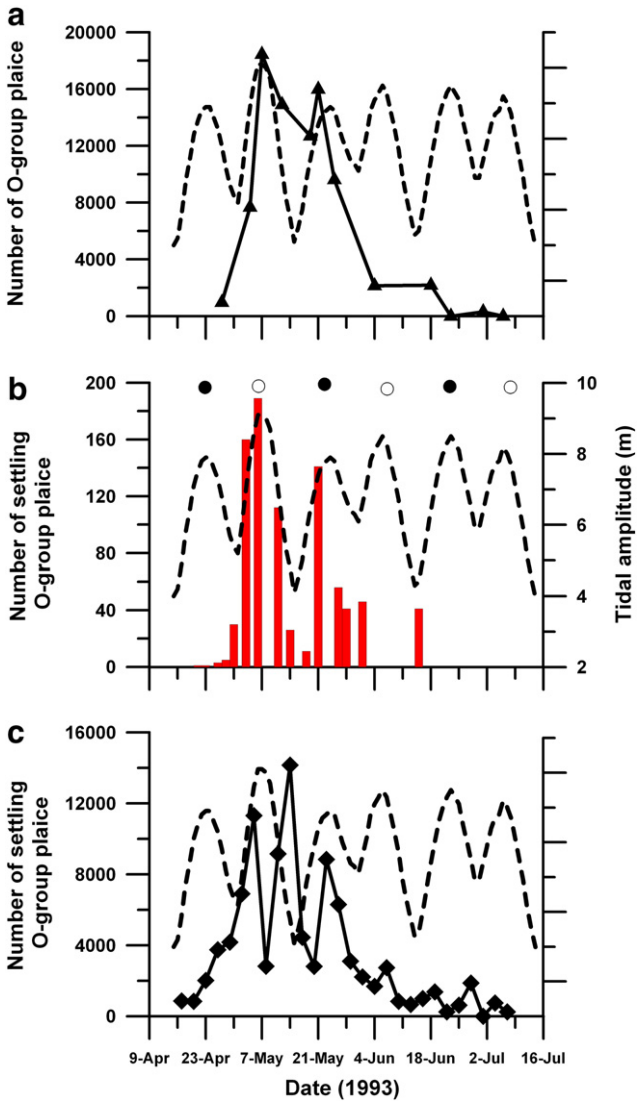

Fig. 2. Numbers of juvenile plaice settling in Port Erin Bay during the summer of 1993, a. sampled by 1.5 and 2 m beam trawl, b. sampled by push net (12–17 mm fish), and c. metamorphosis date sub-cohorts, identified by counts of primary otolith increments. Dashed lines = tidal amplitude; open circles = full moon; and closed circles = new moon.

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samples. The majority of newly settled fish arrived between the end of April and the beginning of May. A second peak in settlement occurred at the end of May, and there was some continued settlement throughout the summer. As with the beam trawl samples, both peaks in abundance coincided with spring tides.

The age distributions calculated from otolith increment counts (post-metamorphosis/settlement) indicated that there were three main sub-cohorts and a group of late settlers during the 1993 season in Port Erin Bay (Fig. 2c). The first sub-cohort was composed of fish metamorphosing between 19th April and 9th May and fish in the second sub-cohort metamorphosed between 10th and 18th May. The first sub-cohort contained more fish (30,957 individuals) than the second sub-cohort (20,709 individuals). The third sub-cohort contained 21,089 individuals and these fish metamorphosed between 19th May and 2nd June. There were also late settlers, which did not form a sub-cohort *per se* because they metamorphosed in low numbers over a long time period. In addition, most of these fish were sampled in September and October and their otoliths were more difficult to read, leading to potential errors in ageing and allocation to sub-cohort.

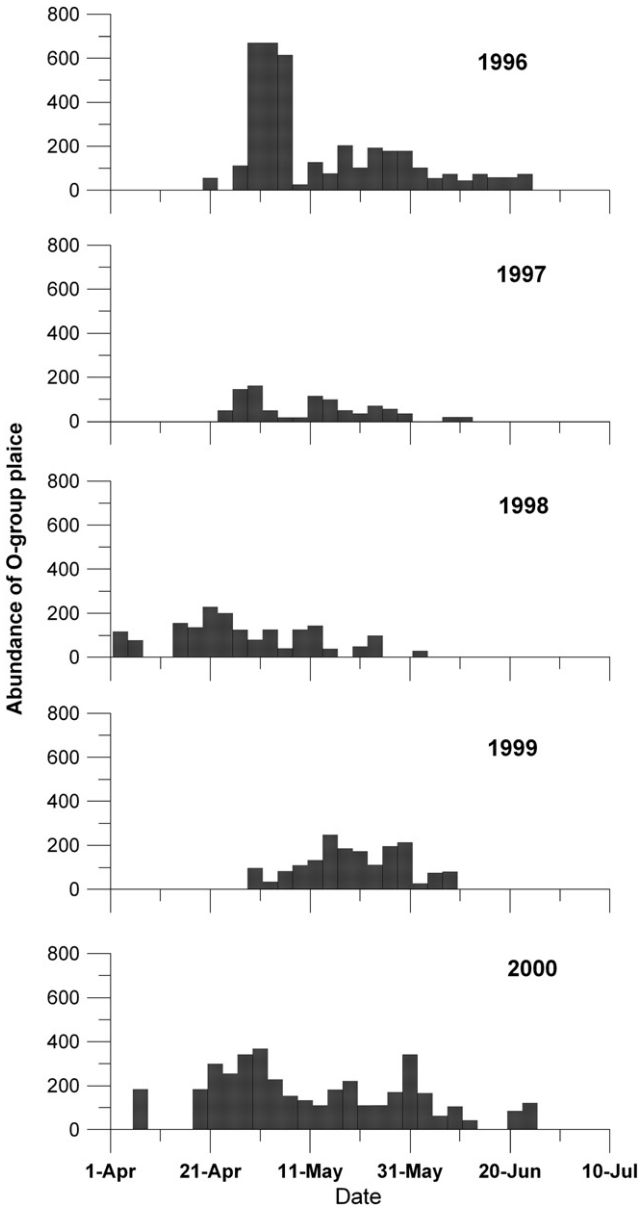

Fig. 3. Settlement patterns of 0-group plaice in Port Erin Bay (Isle of Man) for the period 1996 to 2000.

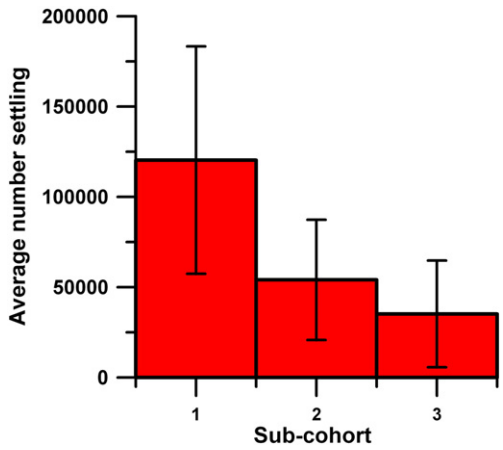

Fig. 4. Settlement patterns of 0-group plaice sub-cohorts in Port Erin Bay (Isle of Man), average numbers of individuals settling in each sub-cohort over the years 1996–2000 (±SD).

The peaks in metamorphosis, as determined from otolith increment counts, occurred at approximately one-week intervals, coinciding with both spring and neap tides. The trawl data indicated that settlement occurred primarily at spring tides. However, the trawls are, by definition, sampling fish which are on the bottom and have settled. The otolith data primarily indicate the timing of metamorphosis rather than the time of settlement and if fish metamorphose prior to settlement there can be a difference in the patterns produced from the two data sets.

3.1.2. Inter-annual variations in settlement and sub-cohort patterns

Settlement patterns varied in both timing and duration between the study years (1996–2000) (Fig. 3). Phenology of settlement differed between years, beginning as early as the 2nd of April (1998) or as late as the 26th of April (1999) and ending as early as the 31st of May (1998) or as late as the 21st of June (1996). Early settlement peaks preceding the main period of migration were identified in 1998 and 2000. Three main peaks of settlement were identified in each of the years, corresponding to sub-cohorts. Late settlers were identified in 1997, 1998, 1999 and 2000. The first settlement sub-cohort was consistently the largest, with decreasing numbers of fish in subsequent sub-cohorts (Fig. 4). The highest peak density of settlement was recorded in 1996, exceeding 600 individuals 1000 m^{-2}.

Back-calculated settlement dates (from otolith increment counts) did not reveal any correspondence between the tidal cycle and settlement (Fig. 5).

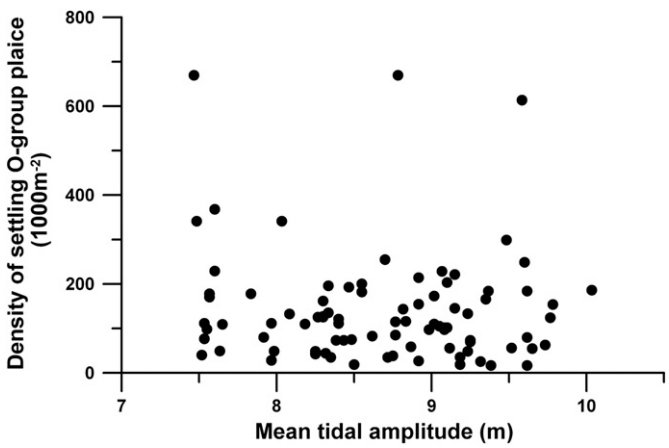

Fig. 5. Relationship between mean tidal amplitude (Laver's Liverpool and Irish Sea Tide Tables) and densities of settling plaice over the years 1996 to 2000.

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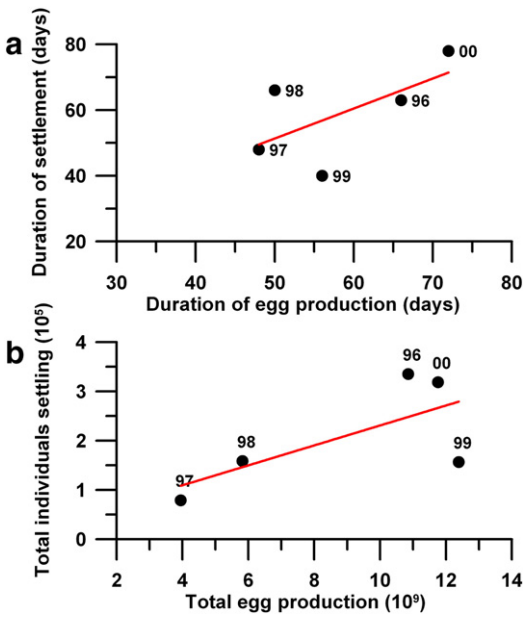

Fig. 6. Relationship between plaice egg production on the west side of the Isle of Man and settlement in Port Erin Bay for the years 1996 to 2000. a. Duration of egg production versus duration of settlement and b. total egg production versus total numbers settling.

Durations of settlement generally corresponded to the duration of egg production (Fig. 6a). The total numbers of individuals settling in Port Erin Bay were positively correlated with the total annual egg production, although in 1999 settlement appeared much lower than expected (Fig. 6b). Larval stage duration was particularly long and the duration of the settlement period was also the shortest in 1999.

3.2. Sub-cohort dynamics

3.2.1. Growth

The intensive sampling in 1993 picked out size differences between sub-cohorts (Fig. 7a), which were maintained throughout the summer (Fig. 7b). By late September the fish in all three sub-cohorts were approximately of the same mean length, possibly as a consequence of emigration of the larger juveniles. Growth rates of fish from each assigned sub-cohort were calculated from length-at-age data over the entire season. The growth trajectories of the sub-cohorts were similar and variability in length increased with age in each sub-cohort (Fig. 7a, Table 1). Overall, the growth of the fish in the three main sub-cohorts was faster than the growth of the late settlers. Growth rates were not age-dependent, and variability in growth rate ranged from 0.3 to 0.8 mm day^{-1} among all the fish (Fig. 7c).

Growth rate in juvenile plaice is generally assumed to be a consequence of the ambient thermal regime (Fonds et al., 1992). The maximum water temperatures in Port Erin Bay usually occur in mid August and the maximum growth rates should have occurred then. The theoretical (thermally driven) mean length of each sub-cohort was calculated for the 1993 data using the Port Erin Bay sea surface temperature series and compared with the observed lengths. In all three sub-cohorts the observed mean length exceeded the theoretical mean length between the beginning of June and mid July (Fig. 7d). Thereafter the actual mean length was less than the theoretical mean length, and the deviation increased with time. Emigration of larger fish may have artificially exaggerated the magnitude of the deviation.

Growth rates were inversely related to the numbers of settlers suggesting an effect of density on growth rate (Fig. 8a). Individual sub-cohort growth rates varied across the years between 0.32 m day^{-1} (2000) and 1.16 mm day^{-1} (1999) (Fig. 8b). As in 1993, the longer-term data series showed higher growth rates in later sub-cohorts as a

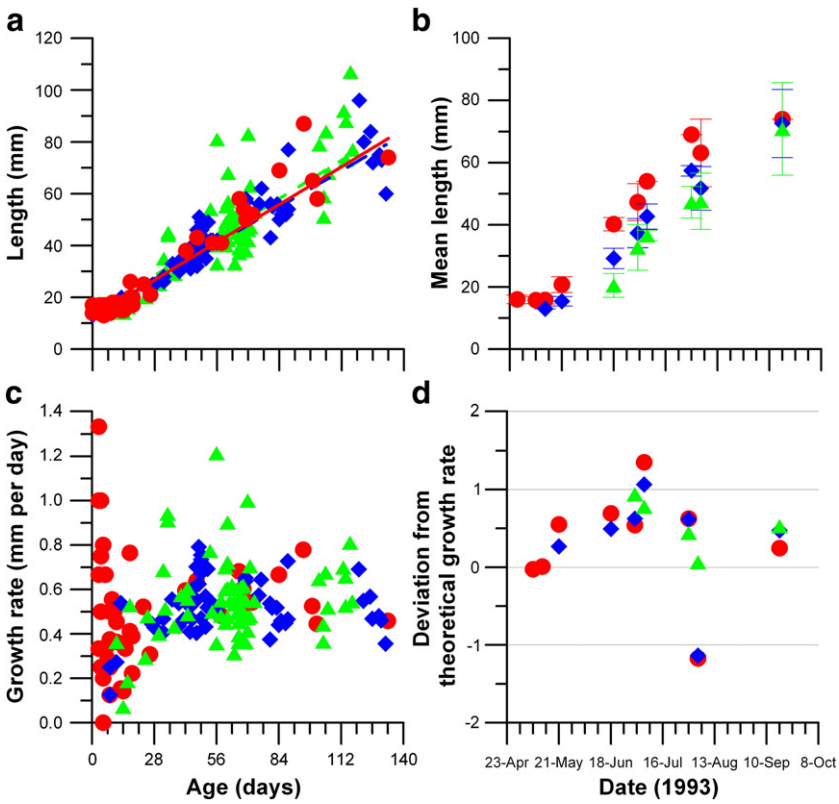

Fig. 7. Length-at-age (post-metamorphosis) and growth rates of individual plaice from beam trawl samples in 1993. a. Length versus age, b. mean length ± SD of the three sub-cohorts on each sampling date, c. growth rate of each sub-cohort with age and d. deviation from the theoretical growth rate based on water temperature (see Fonds et al., 1992) on each sampling dates for each of the three sub-cohorts. Circles = first sub-cohort; diamonds = second sub-cohort, and triangles = third sub-cohort.

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Table 1
Linear growth equations for the three sub-cohorts and late settlers in Port Erin Bay during the summer and autumn of 1993.

Sub-cohort 1:	TL = 0.52 * age + 12.25 (R^2 = 0.94; F(1;163) = 2514; P ≪ 0.01)
Sub-cohort 2[a]:	TL = 0.50 * age + 12.79 (R^2 = 0.89; F(1;209) = 1733; P ≪ 0.01)
Sub-cohort 3[a]:	TL = 0.53 * age + 13.65 (R^2 = 0.67; F(1;220) = 461; P ≪ 0.01)
Late settlers:	TL = 0.49 * age + 17.36 (R^2 = 0.82; F(1;1080) = 110,067; P ≪ 0.01)

TL = total length; and age = days from metamorphosis.
[a] Significantly different growth rates (Tukey pairwise comparison, q = 2.11, P ≪ 0.01).

general trend. As expected, mean sub-cohort growth rates over the five-year period were positively correlated with the weighted mean seawater temperature (Fig. 8c).

3.2.2. Mortality

The instantaneous mortality rate (EMR, Nash and Geffen, 2000) calculated from the catch data was 0.044, on average, fluctuating by a factor of 5 between years for the period 1996 to 2000 (Table 2).

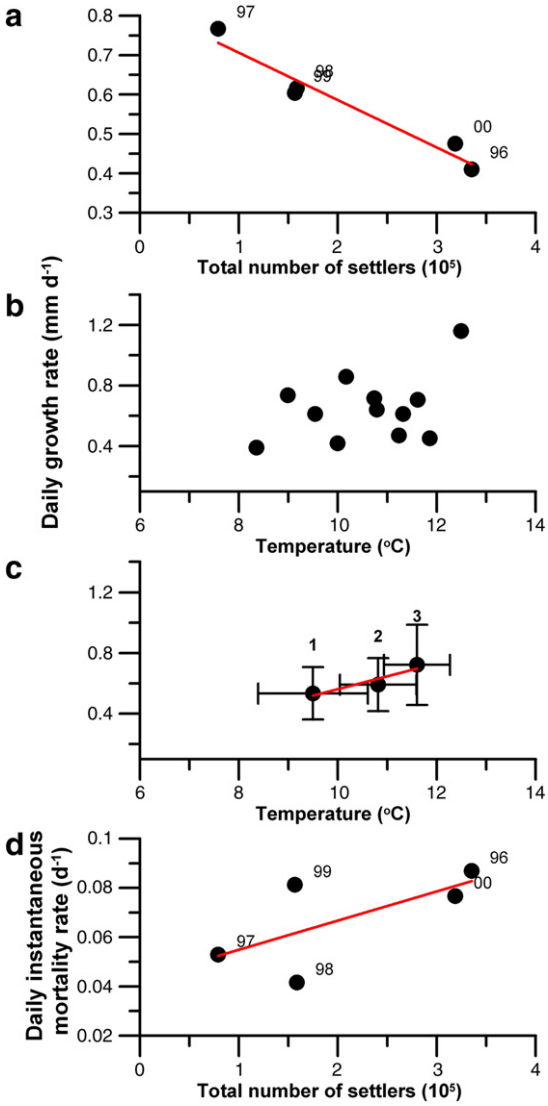

Fig. 8. Plaice sub-cohort growth and mortality on the Port Erin Bay nursery ground from 1996 to 2000. a. Annual growth rates in relation to numbers of settling individuals, b. daily growth rates of each sub-cohort in relation to the mean temperature during the settlement period, c. mean growth rate in relation to the mean temperature for each sub-cohort, and d. daily instantaneous mortality rate in relation to numbers of settlers each year.

Table 2
Daily instantaneous mortality rates for sub-cohorts and EMR (settlement as estimated by Nash and Geffen, 2000) of plaice in Port Erin Bay. Sub-cohorts were identified from otolith micro-increments.

Year	Sub-cohort 1	Sub-cohort 2	Sub-cohort 3	EMR
1993	0.0456	0.0242	0.0263	0.0248
1996	0.2708	0.0749	0.0031	0.0862
1997	0.0837	0.0510	0.1224	0.0234
1998	0.0847	0.1386	0.1390	0.0145
1999	0.0336	0.0560	0.1099	0.0624
2000	0.1619	0.0416	0.0405	0.0320
Average [1996–2000]	0.1269	0.0724	0.0830	0.0437
(SD)	(0.0926)	(0.0389)	(0.0583)	(0.0298)

Mortality rates calculated from otolith data were much higher. On average, between 1996 and 2000 the first sub-cohort suffered the highest mortality, however, the pattern was quite variable and higher mortalities were calculated for the third sub-cohort in 1997–1999. Annual instantaneous mortality rates increased with fish density (Fig 8d).

Shrimp (*C. crangon*) and crabs are reported to be major predators of newly settled plaice on other nursery grounds such as the Wadden Sea (see van der Veer and Bergman, 1987). *C. crangon* and crab abundances varied annually in Port Erin Bay, but rarely coincided with plaice settlement (Fig. 9). Fish greater than 30 mm are relatively immune to predation from crabs and shrimp (Burrows et al., 2001) and density effects on growth could delay the date when these fish cease to be vulnerable. In fact, the dates on which the latest settling fish within each year reached 30 mm varied by only 17 days across all the years. In 1997 (a weak year class) the fish reached this length the fastest, and in 1996 (strong year class) the fish reached it the latest. It appears that the fish were vulnerable to predation for a longer time frame in 1996, compared to 1997, indicative of density dependent predation as a function of growth, which itself is likely to be density dependent on this nursery ground.

Potential fish predators were only examined in 1996 and 1997 (Fig. 10). A greater density was observed in 1997 versus 1996 during the settlement period. In May the principal potential predator in both years was whiting (*Merlangius merlangus*) with other gadoids such as *Trisopterus esmarkii, Ciliata mustela, Gadus morhua, Pollachius pollachius* and *P. virens* also present.

3.3. Irish Sea plaice population trends

Shifts in environmental conditions can affect early life history processes, resulting in changes in the survival schedules. For the wider Irish Sea plaice population, survival over the first year of life declines with increasing temperature (P ≪ 0.01) (Fig. 11a). The relationship between the size of the spawning population and recruitment (measured by the number of 1 year old plaice) appears to be sensitive to the thermal regime (Fig. 11b). Prior to 1980, the stock recruitment relationships were clearly over compensatory (negative slope) and separated for warm and cold years. Since 1980 there has been no obvious relationship between the size of the spawning population and the level of recruitment, however, the separation between warm and cold years remains.

4. Discussion

The presence of sub-cohorts caused by pulses in settlement can have a profound effect on our perception of juvenile plaice growth and mortality rates, especially where there are differences between the sub-cohorts (Al-Hossaini et al., 1989; Hovenkamp, 1991; Karakiri et al., 1991; Sogard and Able, 1992; Modin and Pihl, 1994; this study). Al-Hossaini et al. (1989) used otolith increment data to identify three sub-cohorts of juvenile plaice in Red Wharf Bay (North Wales, Irish

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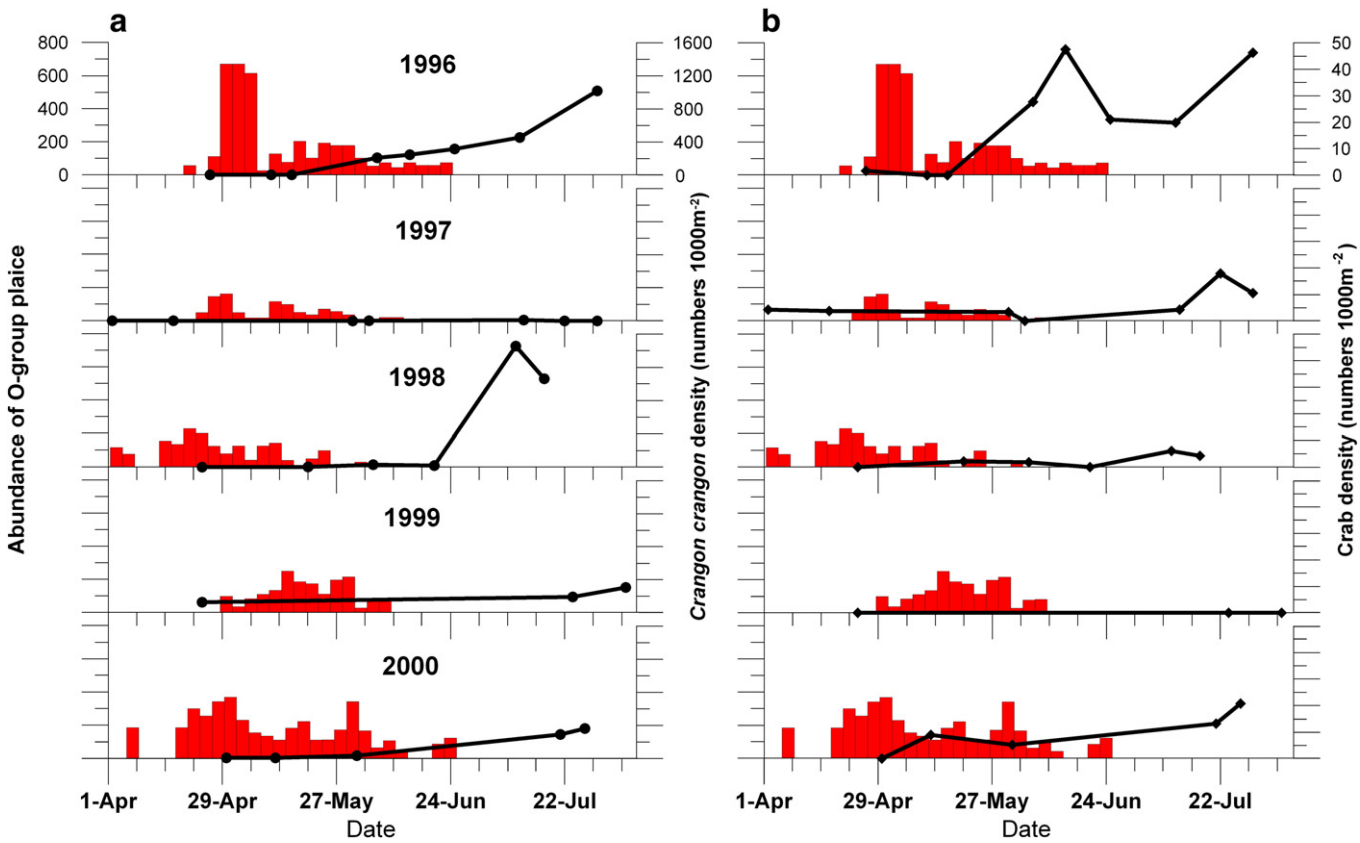

Fig. 9. Settlement patterns of plaice in Port Erin Bay and the abundance of a. shrimp (*Crangon crangon*) and b. crabs on each sampling date.

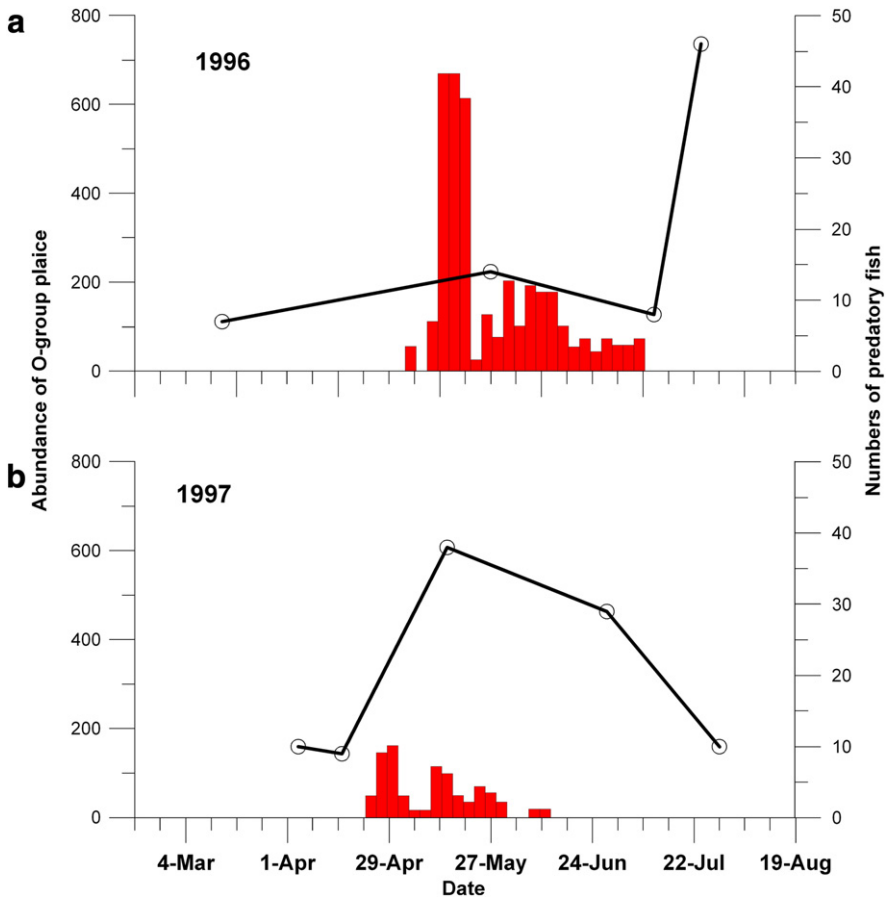

Fig. 10. Abundance of settling 0-group plaice and predatory fish caught by beach seine in shallow water in Port Erin Bay in 1996 and 1997.

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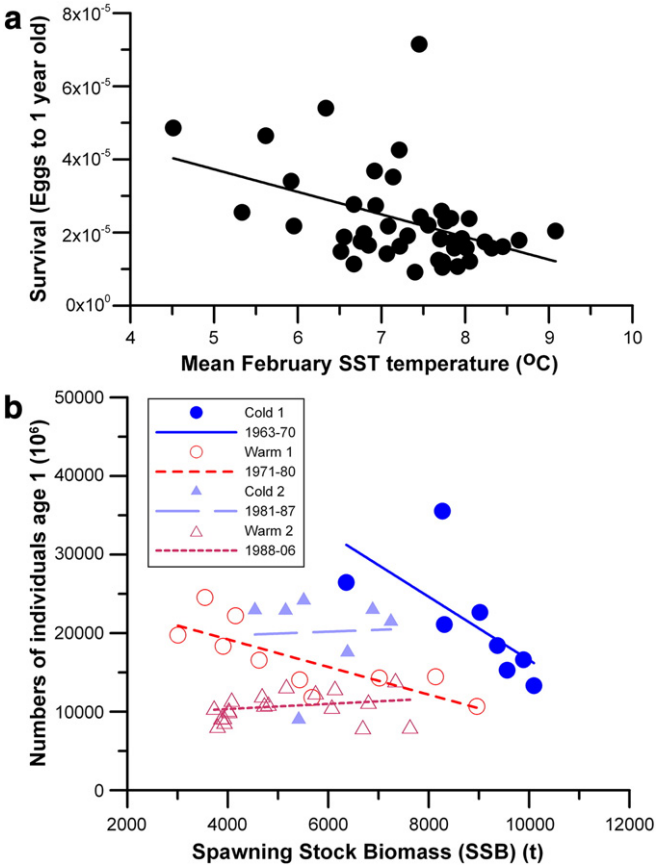

Fig. 11. The plaice (*Pleuronectes platessa*) population in the Irish Sea. a. Proportion surviving from eggs to 1 year old and mean February sea-surface water temperature in Port Erin Bay (Isle of Man, central Irish Sea) and b. relationships between Spawning Stock Biomass (SSB) and the numbers of age 1 individuals for four different thermal regimes (warm 1 and 2 and cold 1 and 2 are as in Nash, 1998 and extended to 2006). In all cases trend lines are shown.

Sea) in 1986 and 1987. In 1986 the third sub-cohort (end of May) was the most abundant, and in 1987 the second sub-cohort was the most abundant. No late settlers were found, but sampling was concentrated during and just after the settlement period and may have missed any late settlers.

4.1. Settlement

Sub-cohorts with different settling dates were identified from catch data which, identified newly settling fish from the presence of individuals 12–17 mm in size (in 1993), and from otolith increment analysis which grouped fish by date of metamorphosis (in 1993 and 1996–2000).

Table 3
Summary of juvenile plaice growth rates determined from otolith increment analyses.

Location	Year	Growth rate (mm day^{-1})	Authority
Red Wharf Bay, Irish Sea	1986	0.50–0.53	Al-Hossaini et al. (1989)
	1987	0.36–0.46	
Gullmar Fjord, Sweden	1991	0.65–0.80	Modin and Pihl (1994)
	1992	0.40–0.70	
Port Erin Bay, Irish Sea	1993	0.49–0.53	This study
	1996	0.39–0.71	
	1997	0.72–0.86	
	1998	0.47–0.64	
	1999	0.45–1.16	
	2000	0.32–0.59	

The settlement peaks as observed directly differed from the identified age sub-cohorts. In 1993 the catch data indicated two settlement peaks: the 27th April–11th May, and the 19th May–1st June, each coinciding with spring tide (full and new moon). These peaks were composed of individuals belonging to different age sub-cohorts that metamorphosed on different dates.

Sub-cohorts, as opposed to peaks in catches of newly settled fish, were defined according to the dates of metamorphosis determined from otolith increment counts. In Port Erin Bay, the otolith analysis from the 1993 fish suggested three main settlement dates between May and June with a slow trickle of newly settling fish thereafter.

The dates of peak settlement determined from otolith increment counts were not related to the lunar cycle. There were three identifiable peaks in the timing of metamorphosis, and fish which metamorphosed on different dates were found combined in the two settlement peaks identified in the catch data. There has always been considerable debate about whether plaice metamorphose and settle in deep or shallow water. Lockwood (1974) suggested that the 0-group plaice settle offshore, before migrating into shallow water. Creutzberg et al. (1978), however, suggested that the end of metamorphosis takes place in shallow water. In Port Erin Bay it is likely that metamorphosis dates are linked to spawning cycles, however, a lunar rhythm seems to be imposed on settlement on the beach. If an individual completes metamorphosis during neap tides, its movement to the beach may be delayed until the next spring tide. However, if the end of metamorphosis coincides with a spring tide, the post-larvae migrate directly into shallow water on the nursery ground. Five percent of the plaice had only 0 or 1 ring after the formation of the first accessory primordium, which means that they settled on the beach just after metamorphosis. By timing settlement with the tidal cycle, metamorphosing plaice may aggregate and settle in large numbers, thus swamping potential predators. Although it has also been suggested that sub-cohorts are the result of fish coming from more distant spawning areas (Al-Hossaini et al., 1989), the close correspondence between egg production and settlement dates in this population suggests that Port Erin Bay is supplied primarily from a local spawning ground (see Nash and Geffen, 1999).

A distinct change in otolith increment width and contrast marks the final stages of eye migration and metamorphosis (Dau, 1994; Modin and Pihl, 1994). The first accessory growth centres develop at the end of metamorphosis, although up to 12% of the fish do not form the first accessory growth centre until 1 to 3 days after metamorphosis (Dau, 1994). Thus otolith analysis can only provide data on the timing of metamorphosis. The frequency of sampling provided reliable catch data to show that the two events (metamorphosis and settlement on the beach) did not always coincide in Port Erin Bay.

Three main sub-cohorts were consistently identified in Port Erin Bay settlement, and the similarity to observations in Red Wharf Bay (Irish Sea) (Al-Hossaini et al., 1989) and the west coast of Ireland (Allen et al., 2008) suggests it to be a general feature of plaice nursery grounds, at least in these latitudes. The resolution of these groups was highly variable, which may have been caused by a combination of the magnitude and pattern of egg production. The settlement pulses observed in the juvenile plaice catch data are different from those in other species with pulsed recruitment because each settlement group contains fish of different ages post-metamorphosis.

The first sub-cohort was usually the largest, and the subsequent sub-cohorts were generally smaller. The causes of this are uncertain since the egg production curves tend to be normally distributed over time (see Nash and Geffen, 1999; Armstrong et al., 2001; Fox et al., 2007).

4.2. Growth

Fish in the second and third sub-cohorts experienced higher temperatures and were larger at age than fish in the first sub-cohort.

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Late settlers were smaller at age. The Port Erin Bay results are very similar to those of the Red Wharf Bay (Al-Hossaini et al., 1989) and Torseröd Bay in the Gullmar Fjord, Sweden (Modin and Pihl, 1994) (Table 3). In both bays the highest growth rates occurred in the second sub-cohort, which they attributed to higher temperature. Empirically determined growth models (Fonds et al., 1992) suggest that these late settlers did not grow at their maximal rate with respect to temperature (see Nash et al., 1994a). Interspecific food competition may influence growth in this sub-cohort.

Hovenkamp (1991) demonstrated that late settling plaice can grow up to 3 times as fast as early settlers, thus growing through certain vulnerable sizes in approximately a quarter of the time it would take early settlers. The relative proportions of sub-cohorts in the population influence the estimates of mean growth rates. Similarly, the survival of each sub-cohort may affect the estimated population mortality rate, possibly going part way to explaining interannual variability in estimates of nursery ground mortality rates. The characteristics of the individual sub-cohorts form the overall patterns observed on the nursery grounds and therefore it is important to appreciate the sub-cohort dynamics when interpreting the overall dynamics of 0-group plaice on nursery grounds.

4.3. Mortality

Nursery ground mortality is usually attributed to predation, and density-dependent effects tend to dampen variability in abundance (Bailey, 1994). There may be indirect density effects through growth rate, size-selectivity of predators and thus uneven vulnerability to predation (van der Veer et al., 1997). There may be elevated mortality at metamorphosis (Geffen et al., 2007), however an adaptation to aggregate and swamp predators by coordinating the arrival on the beach is a possible mechanism operating around this period of the life history. This may also explain why the first sub-cohort is larger, if it enables a “high risk” strategy of predator swamping early in the season (van Montfrans et al., 1995). This first sub-cohort often suffered the highest mortality, a phenomenon also found in Red Wharf Bay (Al-Hossaini et al., 1989). The maximum sub-cohort mortality rates recorded in Port Erin Bay (>0.1 day^{-1}) were much higher than those reported for Red Wharf Bay (approximately 0.07 day^{-1}), but were close to the maximum (0.15 day^{-1}) reported by van der Veer (1986) for the Wadden Sea. Both Irish Sea sites showed interannual variability in sub-cohort mortality rates.

Plaice spawn relatively early in the year with settlement often occurring prior to the influx of other 0- and 1-group fish i.e. competitors or predators returning to shallow water after the winter. One of the species identified as a principal predator of very young plaice is the shrimp *C. crangon* (van der Veer, 1986; van der Veer and Bergman, 1987). Shrimps with a carapace length of >30 mm can prey on the young plaice up to a size of approximately 30 mm (van der Veer and Bergman, 1987; Gibson et al., 1995; Burrows et al., 2001), and they overwinter in the vicinity of or on the nursery grounds and return to the settlement area in early spring (van der Veer and Bergman, 1987). In extremely cold winters the overwintering shrimp either die off or move to much deeper water, thus releasing predation pressure on the settling plaice (cf. van der Veer and Bergman, 1987) with the biggest impact being that of reduced mortality of early settlers, and especially the first sub-cohort. Cold winters tend to produce large year classes of plaice, but this could simply be the result of greater survival in the pelagic phase (van der Veer and Bergman, 1987). High settlement abundance tends to be associated with large year classes (Nash and Geffen, 2000) but it is hard to quantify the contribution that a change in sub-cohort mortality rate makes to the eventual numbers of individuals leaving a nursery ground. As the season progresses, the shrimps on the nursery ground grow (Oh et al., 1999; Burrows et al., 2001) and more predators also populate the area. Thus later settling sub-cohorts will become vulnerable to higher predation.

Gibson et al. (2002) indicate that settlement is initially in the shallowest water and later settlers (sub-cohorts) will settle in deeper water due to density constraints. They also argue that predation pressure or risk of mortality is higher in deeper water, thus the expectancy would be for mortality to be higher in later cohorts. In Port Erin Bay the second sub-cohort always experienced the lowest mortality rate.

The extent of bird predation on juvenile plaice has not been adequately determined. However, there is a strong possibility that avian predation on settling individuals during the period before early July is not a significant source of mortality since parental birds are not feeding chicks at this time and the birds tend to predate fish in deeper water (see van der Veer et al., 1997). However, this may be an artefact of when the majority of sampling is undertaken to determine avian predation. Cormorants in the Wadden Sea predated plaice as small as 40 mm in September, so small plaice are potentially vulnerable to avian predation. However, the mortality of sub-cohorts resulting from size selective predation during this latter part of the year is unknown.

A number of species of fish predate 0-group plaice (van der Veer et al., 1997) along with cannibalism also being reported for two Irish Sea nursery grounds (Riley and Corlett, 1966; Macer, 1967). The 1996–1997 data indicated that there were relatively few 1+ plaice in the bay (de la Rosa, 1999) thus suggesting cannibalism was probably relatively low. The number of 1-group plaice overwintering on the nursery is usually small, but when larger year classes occur there is a tendency for more individuals to remain in the nursery ground to overwinter (Nash et al., 1994a,b). In these situations there is the possibility that newly settling plaice in the year following a large year class could suffer higher levels of cannibalism. Ellis and Gibson (1995) showed a negative size selection by piscine predators on the 0-group flatfishes, primarily concentrated on the smaller size classes. Because only the larger fish predators tended to predate the flatfish there is the potential for interannual variability in predation levels based on the relative overlap in sizes and abundances of both predator and prey. Ellis and Gibson's (1995) paper concentrated on smaller predators, and faster growing individuals could outgrow a gadoid predation window when they reached a size refuge at approximately 42 mm. This has implications for sub-cohort dynamics as predation levels will be a combination of plaice settlement date (plus cohort number), predator arrival dates and plaice growth rates.

The population dynamics of separate sub-cohorts provide an insight into the factors that may be affecting the overall juvenile cohort survival and therefore the potential for recruitment in each of the years examined. In general, growth rates in each of the years were of the same order of magnitude as in other studies (Table 3). The relationship between the growth rate and the total abundance of juveniles on the nursery ground appeared strong at high densities. This supports previous reports of density dependent processes in a nursery ground environment. At lower abundance the impact of density dependent factors would be lower and therefore other factors could potentially influence the growth rate, causing a greater degree of fluctuation.

4.4. Environmental change and survival in early life history stages

Global environment change effects are visible in the plaice population dynamics, with temperature dependent nursery ground processes as one of the contributing mechanisms. Temperature affects both growth rate and predator abundance. Nash and Geffen (2000) found no relationships between temperature and juvenile plaice instantaneous mortality rates in Port Erin Bay. However, when viewed from the perspective of the longer time series, the period of study was actually in a sustained period of warmer conditions, thus the effect of ‘cold’ winters was not visible in that data set. At the scale of the wider

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Irish Sea plaice population, recruitment (which is an index of survival through the first year of life) is generally higher at colder temperatures. This response is also reported by Fox et al. (2000) and seen in other species, e.g. cod (*G. morhua*) at the southern end of its latitudinal range (Drinkwater, 2005 and references therein). Superimposed on this relationship is the longer term pattern visible over the last four decades. Differences between cold and warm periods were also demonstrated in the Paulik diagram models of Irish Sea plaice given in Nash (1998). These patterns are likely to be the results of combined effects of global environmental changes and direct effects of the fishery.

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